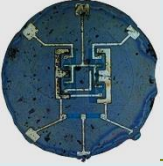


Semiconductor Physics and Devices

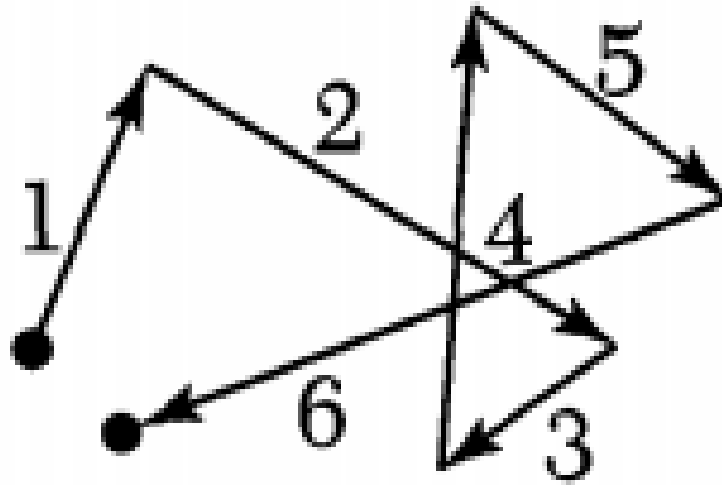
Lecture 7 Conductivity and Mobility

ETE-301



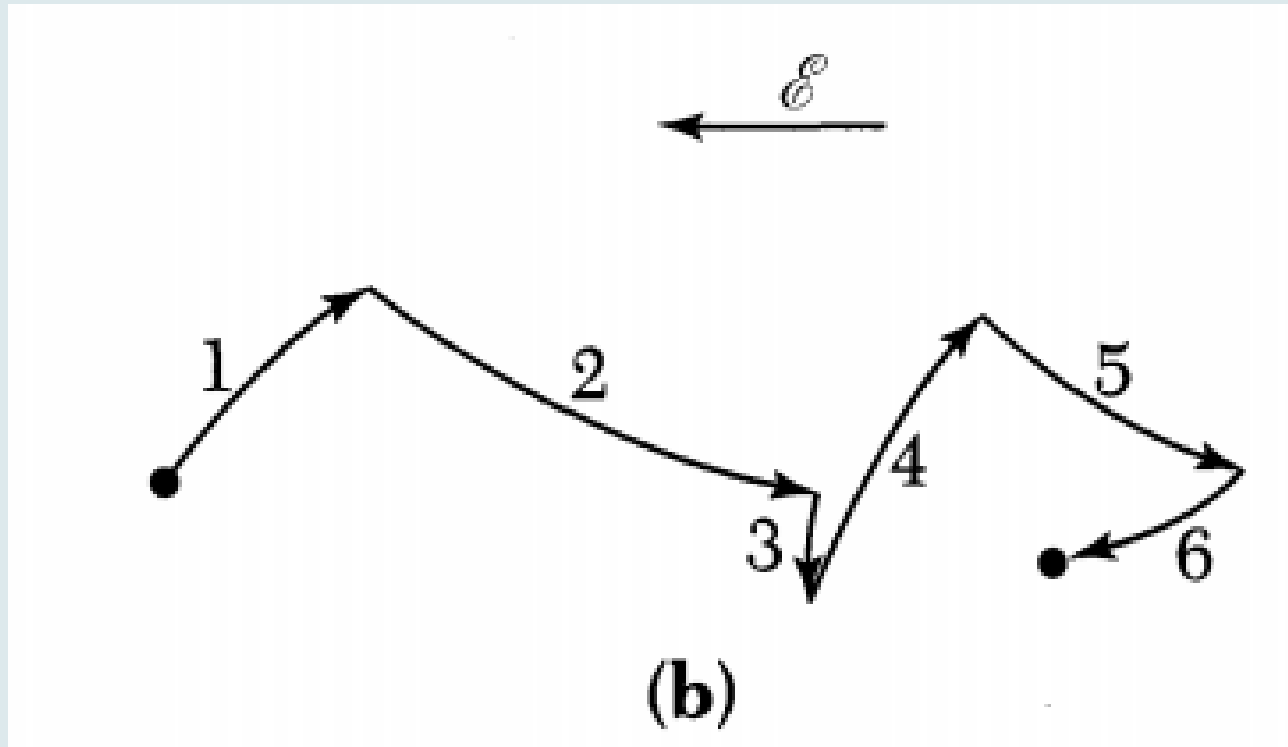
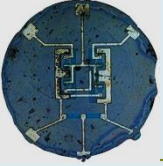
Conductivity and Mobility

$$\mathcal{E} = 0$$

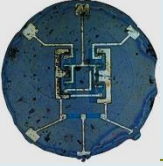


(a)

(a) Random thermal motion.

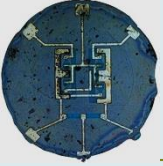


(b) Combined motion due to random thermal motion and an applied electric field.



If an electric field $\mathcal{E}_x$ is applied in the x -direction, each electron experiences a net force $-q\mathcal{E}_x$ from the field. This force may be insufficient to alter appreciably the random path of an individual electron; the effect when averaged over all the electrons, however, is a net motion of the group in the $-x$ -direction. If $\mathbf{p}_x$ is the x -component of the total momentum of the group, the force of the field on the n electrons/cm³ is

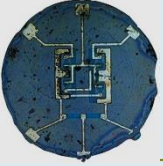
$$-nq\mathcal{E}_x = \left. \frac{d\mathbf{p}_x}{dt} \right|_{\text{field}} \dots\dots\dots(1)$$



To find the total rate of momentum change from collisions, we must investigate the collision probabilities more closely. If the collisions are truly random, there will be a constant probability of collision at any time for each electron. Let us consider a group of N_0 electrons at time $t = 0$ and define $N(t)$ as the number of electrons that *have not* undergone a collision by time t . The rate of decrease in $N(t)$ at any time t is proportional to the number left unscattered at t ,

$$-\frac{dN(t)}{dt} = \frac{1}{\bar{t}} N(t) \dots\dots\dots(2)$$

where $\bar{t}^{-1}$ is a constant of proportionality.



The solution of equation (2) is an exponential function

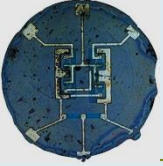
$$N(t) = N_0 e^{-t/\bar{t}} \dots\dots\dots(3)$$

and $\bar{t}$ represents the mean time between scattering events, $\bar{t}$ called the *mean free time*. The probability that any electron has a collision in the time interval dt is $dt/\bar{t}$. Thus the differential change in p_x due to collisions in time dt is

$$dp_x = -p_x \frac{dt}{\bar{t}} \dots\dots\dots(4)$$

The rate of change of p_x due to the decelerating effect of collisions is

$$\left. \frac{dp_x}{dt} \right|_{\text{collisions}} = -\frac{p_x}{\bar{t}} \dots\dots\dots(5)$$



The sum of acceleration and de-acceleration must be zero. Taking the sum of equ. 1 and 5, we have

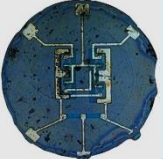
$$-\frac{p_x}{t} - nq\mathcal{E}_x = 0 \quad \dots\dots\dots(6)$$

The average momentum per electron is

$$\langle p_x \rangle = \frac{p_x}{n} = -q\bar{t}\mathcal{E}_x \quad \dots\dots\dots(7)$$

where the angular brackets indicate an average over the entire group of electrons. As expected for steady state, Eq. (7) indicates that the electrons have *on the average* a constant net velocity in the negative x -direction:

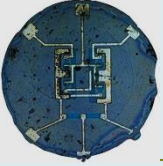
$$\langle v_x \rangle = \frac{\langle p_x \rangle}{m_n^*} = -\frac{q\bar{t}}{m_n^*}\mathcal{E}_x \quad \dots\dots\dots(8)$$



Actually, the individual electrons move in many directions by thermal motion during a given time period, but Eq. (8) tells us the *net drift* of an average electron in response to the electric field. The drift speed described by Eq. (8) is usually much smaller than the random speed due to thermal motion V_{th} .

The current density resulting from this net drift is just the number of electrons crossing a unit area per unit time ($n\langle v_x \rangle$) multiplied by the charge on the electron ($-q$):

$$J_x = -qn\langle v_x \rangle$$
$$\frac{\text{ampere}}{\text{cm}^2} = \frac{\text{coulomb}}{\text{electron}} \cdot \frac{\text{electrons}}{\text{cm}^3} \cdot \frac{\text{cm}}{\text{s}}$$



Using Eq. (8) for the average velocity, we obtain

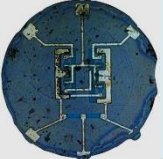
$$J_x = \frac{nq^2\bar{t}}{m_n^*} \mathcal{E}_x \quad \text{.....(9)}$$

Thus the current density is proportional to the electric field, as we expect from Ohm's law:

$$J_x = \sigma \mathcal{E}_x, \quad \text{where } \sigma \equiv \frac{nq^2\bar{t}}{m_n^*} \quad \text{.....(10)}$$

The conductivity $\sigma(\Omega\text{-cm})^{-1}$ can be written

$$\sigma = qn\mu_n, \quad \text{where } \mu_n \equiv \frac{q\bar{t}}{m_n^*} \quad \text{.....(11)}$$



The quantity μ_n , called the *electron mobility*, describes the ease with which electrons drift in the material. Mobility is a very important quantity in characterizing semiconductor materials and in device development.

$$\mu_n = -\frac{\langle \mathbf{v}_x \rangle}{\mathcal{E}_x} \dots\dots\dots(12)$$

The current density can be written in terms of mobility as

$$J_x = qn\mu_n\mathcal{E}_x \dots\dots\dots(13)$$

This derivation has been based on the assumption that the current is carried primarily by electrons. For hole conduction we change n to p , $-q$ to $+q$, and μ_n to μ_p , where $\mu_p = +\langle \mathbf{v}_x \rangle / \mathcal{E}_x$ is the mobility for holes. If both electrons and holes participate, we must modify Eq. (12) to

$$J_x = q(n\mu_n + p\mu_p)\mathcal{E}_x = \sigma\mathcal{E}_x \dots\dots\dots(14)$$